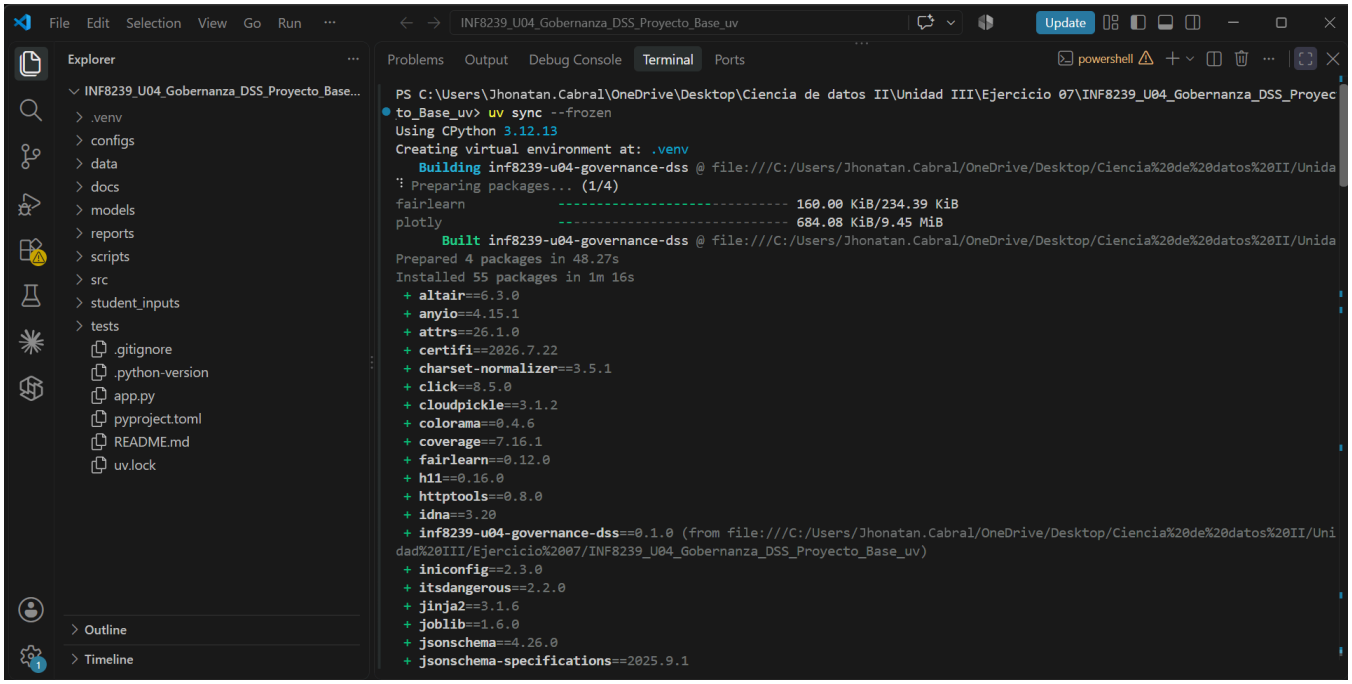
Anexo de evidencias · Ejercicio 07

INF-8239 Ciencia de Datos II, Unidad 04 (U04.E07)

Estudiante	Jhonatan Cabral (BE2322)
Informe	U04_E07_Jhonatan Cabral.pdf
Qué contiene	Capturas de la ejecución de LAB11 en mi equipo (E1–E3) y la salida de consola de LAB12 y E07 (E4). Cada evidencia indica qué se realiza y qué parte del mandato respalda.

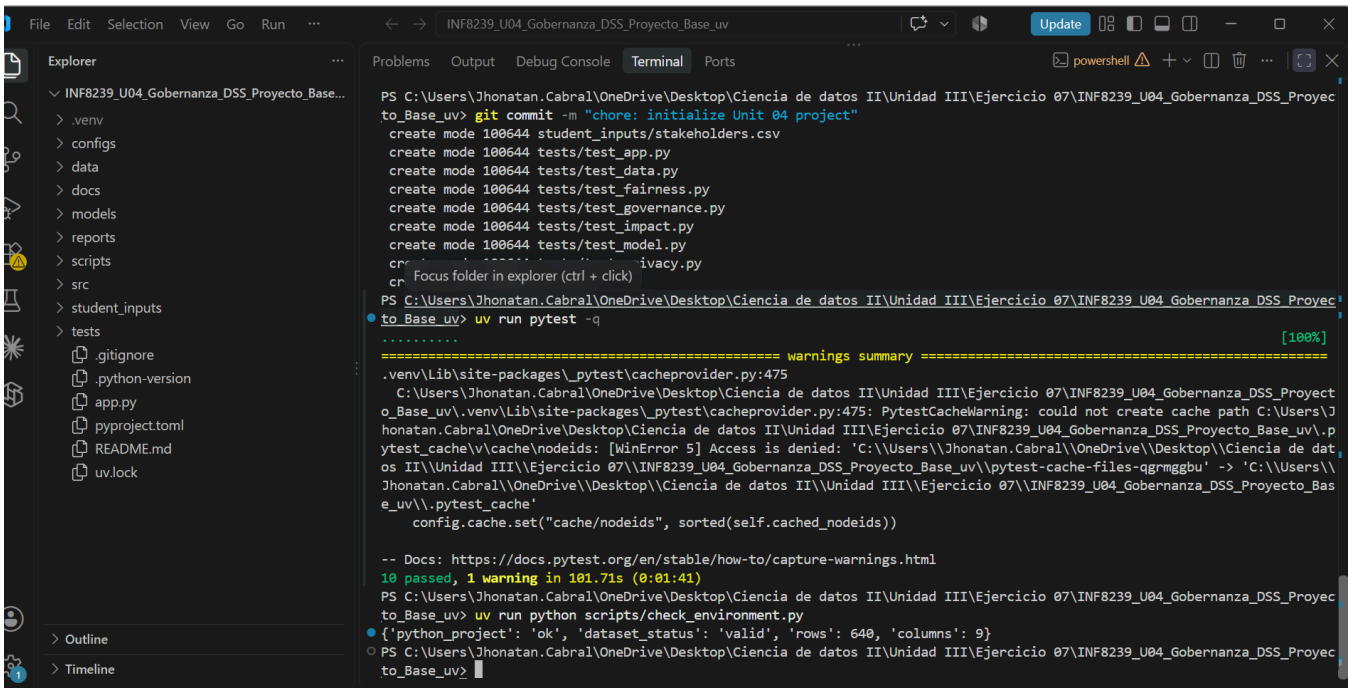
Capturas de LAB11 · Evaluación de impacto ético

Ejecuté LAB11 en VS Code el 30 de octubre de 2026, en una copia local del proyecto (Unidad III/Ejercicio 07). Después la integré al repositorio del curso, en Unidad IV/Ejercicio 07. Solo recorté la barra de estado inferior, que mostraba datos de una cuenta corporativa; el terminal no se modificó.



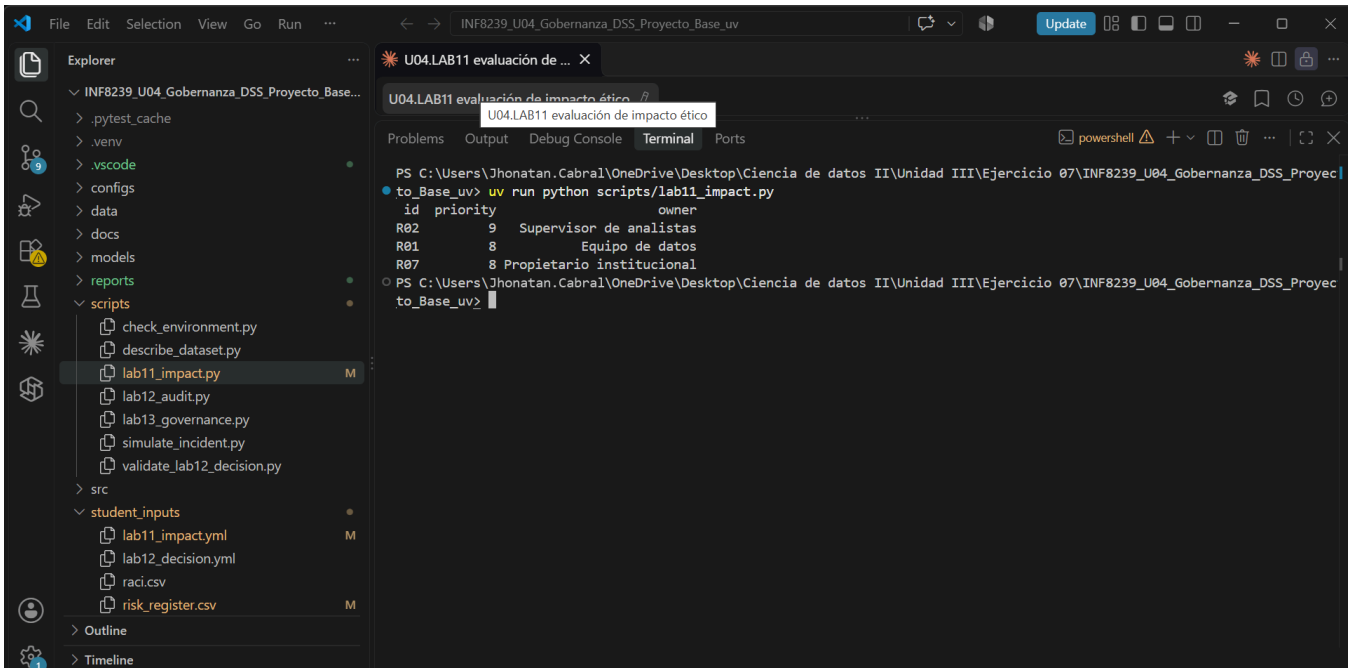
E1. Instalación del entorno reproducible (uv sync --frozen)

Qué se realiza: uv crea el entorno aislado *.venv* con CPython 3.12.13 e instala 55 paquetes con las versiones exactas de *uv.lock*, entre ellos fairlearn 0.12.0 y el propio proyecto. **Mandato:** respalda las pruebas y la trazabilidad, porque cualquier persona obtiene el mismo entorno con el mismo comando.



E2. Versionado inicial, pruebas y contrato de datos

Qué se realiza: el commit inicial (*chore: initialize Unit 04 project*) registra el punto de partida. *pytest* da 10 pruebas aprobadas; el aviso *PytestCacheWarning* se debe a OneDrive y no es un fallo. *check_environment.py* confirma *dataset_status: valid*, con 640 filas y 9 columnas. **Mandato:** pruebas verificables. El contrato técnico se cumple, pero eso no demuestra que los datos sirvan para una decisión real.



E3. Priorización del registro de riesgos (lab11_impact.py)

Qué se realiza: el script valida la ficha de propósito, los interesados y los 8 riesgos, calcula la prioridad (probabilidad × impacto) y genera la evidencia en *reports/lab11/*. Los tres riesgos prioritarios son R02 (9, supervisor de analistas), R01 (8, equipo de datos) y R07 (8, propietario institucional). *lab11_impact.py* aparece modificado (M) por la corrección del error de sintaxis. **Mandato:** registro de riesgos con responsables (NIST MAP).

E4. Salida de consola de LAB12 y E07

Qué se realiza: se ejecutan las pruebas, la comprobación del entorno, la descripción del dataset, la validación de la decisión de LAB12, el análisis de mitigación de E07 y la revisión posterior (soporte por clase y validación temporal de la alternativa elegida). La salida es texto copiado sin cambios de *reports/e07/run_log.txt*, que se puede reproducir con los mismos comandos. **Mandato:** auditoría y mitigación con evidencia editable, no solo capturas.

```
$ uv run pytest -q -p no:cacheprovider
..............                                                           [100%]
14 passed in 5.61s

$ uv run python scripts/check_environment.py
{'python_project': 'ok', 'dataset_status': 'valid', 'rows': 640, 'columns': 9}

$ uv run python scripts/describe_dataset.py
shape (640, 9)
target {1: 0.552, 0: 0.448}
groups {'G1': 396, 'G2': 244}
train_end 2026-09-24 00:00:00
test_start 2026-09-25 00:00:00
excluded_from_model ['case_id', 'received_date', 'protected_group', 'target']

$ uv run python scripts/validate_lab12_decision.py
decision_status: documented

$ uv run python scripts/e07_mitigation.py
          variant  accuracy  recall   fpr  selection_rate  recall_gap  fpr_gap  recall_G1  fpr_G1  recall_G2  fpr_G2  meets_gates
    rules_no_model     0.681   0.859 0.559           0.731       0.084    0.087      0.898   0.538      0.814   0.625         True
    logistic_t0.50     0.719   0.717 0.279           0.531       0.138    0.125      0.653   0.250      0.791   0.375         True
    logistic_t0.45     0.712   0.739 0.324           0.562       0.140    0.149      0.673   0.288      0.814   0.438         True
    logistic_t0.40     0.731   0.848 0.426           0.669       0.067    0.096      0.816   0.404      0.884   0.500         True
expgrad_tpr_parity     0.744   0.728 0.235           0.519       0.030    0.101      0.714   0.212      0.744   0.312         True
{
  "logistic_t0.50": {
    "recall_gap": {
      "p2.5": 0.010827498712249885,
      "p50": 0.137097321464213,
      "p97.5": 0.3139834223498571
    },
```

```
    "fpr_gap": {
      "p2.5": 0.00510160843998617,
      "p50": 0.12718638418526435,
      "p97.5": 0.3975362685931603
    },
    "recall_G1": {
      "p2.5": 0.5217391304347826,
      "p50": 0.6530612244897959,
      "p97.5": 0.7843137254901961
    }
  },
  "expgrad_tpr_parity": {
    "recall_gap": {
      "p2.5": 0.0022725984333127613,
      "p50": 0.06405221588415977,
      "p97.5": 0.21274104977891867
    },
    "fpr_gap": {
      "p2.5": 0.004799107142857154,
      "p50": 0.11542535853976534,
      "p97.5": 0.37341979828681954
    },
    "recall_G1": {
      "p2.5": 0.5918135435992579,
      "p50": 0.717391304347826,
      "p97.5": 0.8421310629514963
    }
  }
}
```


	count	accuracy	selection_rate	recall	precision	fpr
sensitive_feature_0						
G1 | 18-29	18.0	0.833	0.278	0.750	0.600	0.143
G1 | 30-44	24.0	0.750	0.500	0.688	0.917	0.125
G1 | 45-59	35.0	0.686	0.400	0.588	0.714	0.222
G1 | 60+	24.0	0.792	0.625	0.917	0.733	0.333
G2 | 18-29	11.0	0.909	0.636	0.875	1.000	0.000
G2 | 30-44	14.0	0.500	0.500	0.500	0.857	0.500
G2 | 45-59	8.0	1.000	0.750	1.000	1.000	0.000
G2 | 60+	26.0	0.692	0.654	0.765	0.765	0.444


```
{'training_seconds': {'logistic': 0.058, 'expgrad_tpr_parity': 0.404}}
```


```
$ uv run python scripts/e07_review.py
```

Support by class (test):

	variant	level	n	n_pos	n_neg	recall	recall_ci_low	recall_ci_high	recall_supported	fpr	fpr_supported
group											
G1	logistic_t0.50	group	101	49	52	0.653	0.513	0.771	True	0.250	True
G2	logistic_t0.50	group	59	43	16	0.791	0.648	0.886	True	0.375	False
G1 | 18-29	logistic_t0.50	intersectional	18	4	14	0.750	0.301	0.954	False	0.143	False
G1 | 30-44	logistic_t0.50	intersectional	24	16	8	0.562	0.332	0.769	False	0.125	False
G1 | 45-59	logistic_t0.50	intersectional	35	17	18	0.588	0.360	0.784	False	0.389	False
G1 | 60+	logistic_t0.50	intersectional	24	12	12	0.833	0.552	0.953	False	0.250	False
G2 | 18-29	logistic_t0.50	intersectional	11	8	3	0.875	0.529	0.978	False	0.000	False
G2 | 30-44	logistic_t0.50	intersectional	14	12	2	0.667	0.391	0.862	False	0.500	False
G2 | 45-59	logistic_t0.50	intersectional	8	6	2	1.000	0.610	1.000	False	0.000	False
G2 | 60+	logistic_t0.50	intersectional	26	17	9	0.765	0.527	0.904	False	0.556	False
G1	tree	group	101	49	52	0.633	0.493	0.753	True	0.269	True
G2	tree	group	59	43	16	0.814	0.674	0.903	True	0.438	False
G1 | 18-29	tree	intersectional	18	4	14	0.750	0.301	0.954	False	0.214	False
G1 | 30-44	tree	intersectional	24	16	8	0.625	0.386	0.815	False	0.375	False
G1 | 45-59	tree	intersectional	35	17	18	0.647	0.413	0.827	False	0.278	False
G1 | 60+	tree	intersectional	24	12	12	0.583	0.320	0.807	False	0.250	False
G2 | 18-29	tree	intersectional	11	8	3	1.000	0.676	1.000	False	0.000	False
G2 | 30-44	tree	intersectional	14	12	2	0.833	0.552	0.953	False	1.000	False
G2 | 45-59	tree	intersectional	8	6	2	0.667	0.300	0.903	False	0.000	False
G2 | 60+	tree	intersectional	26	17	9	0.765	0.527	0.904	False	0.556	False
G1	expgrad_tpr_parity	group	101	49	52	0.714	0.576	0.822	True	0.212	True
G2	expgrad_tpr_parity	group	59	43	16	0.744	0.598	0.851	True	0.312	False
G1 | 18-29	expgrad_tpr_parity	intersectional	18	4	14	0.750	0.301	0.954	False	0.143	False
G1 | 30-44	expgrad_tpr_parity	intersectional	24	16	8	0.688	0.444	0.858	False	0.125	False
G1 | 45-59	expgrad_tpr_parity	intersectional	35	17	18	0.588	0.360	0.784	False	0.222	False
G1 | 60+	expgrad_tpr_parity	intersectional	24	12	12	0.917	0.646	0.985	False	0.333	False
G2 | 18-29	expgrad_tpr_parity	intersectional	11	8	3	0.875	0.529	0.978	False	0.000	False
G2 | 30-44	expgrad_tpr_parity	intersectional	14	12	2	0.500	0.254	0.746	False	0.500	False
G2 | 45-59	expgrad_tpr_parity	intersectional	8	6	2	1.000	0.610	1.000	False	0.000	False
G2 | 60+	expgrad_tpr_parity	intersectional	26	17	9	0.765	0.527	0.904	False	0.444	False

Temporal validation (training period only):

fold	valid_from	valid_to	variant	recall	fpr	recall_gap	fpr_gap	recall_G1	pos_G1	pos_G2
1	2026-05-13	2026-06-25	logistic_t0.50	0.833	0.250	0.112	0.312	0.773	22	26
1	2026-05-13	2026-06-25	logistic_t0.40	0.896	0.281	0.059	0.264	0.864	22	26
1	2026-05-13	2026-06-25	expgrad_tpr_parity	0.812	0.250	0.010	0.312	0.818	22	26
1	2026-05-13	2026-06-25	rules_no_model	0.875	0.406	0.021	0.074	0.864	22	26
2	2026-06-26	2026-08-11	logistic_t0.50	0.778	0.343	0.277	0.189	0.636	22	23
2	2026-06-26	2026-08-11	logistic_t0.40	0.867	0.400	0.273	0.220	0.727	22	23
2	2026-06-26	2026-08-11	expgrad_tpr_parity	0.756	0.343	0.144	0.189	0.682	22	23
2	2026-06-26	2026-08-11	rules_no_model	0.911	0.571	0.004	0.052	0.909	22	23
3	2026-08-12	2026-09-24	logistic_t0.50	0.810	0.184	0.106	0.006	0.769	26	16

```
    3 2026-08-12 2026-09-24     logistic_t0.40   0.857 0.289       0.130    0.040      0.808      26      16
    3 2026-08-12 2026-09-24 expgrad_tpr_parity   0.833 0.158       0.168    0.057      0.769      26      16
    3 2026-08-12 2026-09-24     rules_no_model   0.905 0.421       0.053    0.080      0.885      26      16
{
  "pooled_metrics": {
    "logistic_t0.50": {
      "accuracy": 0.7791666666666667,
      "recall": 0.8074074074074075,
      "fpr": 0.2571428571428571,
      "selection_rate": 0.5666666666666667,
      "recall_gap": 0.16373626373626382,
      "fpr_gap": 0.14999999999999997,
      "recall_G1": 0.7285714285714285,
      "fpr_G1": 0.2,
      "recall_G2": 0.8923076923076924,
      "fpr_G2": 0.35
    },
    "logistic_t0.40": {
      "accuracy": 0.7875,
      "recall": 0.8740740740740741,
      "fpr": 0.3238095238095238,
      "selection_rate": 0.6333333333333333,
      "recall_gap": 0.15384615384615385,
      "fpr_gap": 0.16346153846153844,
      "recall_G1": 0.8,
      "fpr_G1": 0.26153846153846155,
      "recall_G2": 0.9538461538461539,
      "fpr_G2": 0.425
    },
    "expgrad_tpr_parity": {
      "accuracy": 0.7791666666666667,
      "recall": 0.8,
      "fpr": 0.24761904761904763,
      "selection_rate": 0.5583333333333333,
      "recall_gap": 0.08901098901098903,
      "fpr_gap": 0.125,
      "recall_G1": 0.7571428571428571,
      "fpr_G1": 0.2,
      "recall_G2": 0.8461538461538461,
      "fpr_G2": 0.325
    },
    "rules_no_model": {
      "accuracy": 0.7375,
      "recall": 0.8962962962962963,
      "fpr": 0.4666666666666667,
      "selection_rate": 0.7083333333333334,
      "recall_gap": 0.02197802197802201,
      "fpr_gap": 0.02692307692307694,
      "recall_G1": 0.8857142857142857,
      "fpr_G1": 0.47692307692307695,
      "recall_G2": 0.9076923076923077,
      "fpr_G2": 0.45
    }
  },
  "expgrad_minus_base": {
    "recall_gap": {
      "p2.5": -0.16658457562043183,
      "p50": -0.06828858437314322,
      "p97.5": 0.030770872865275144,
      "share_below_0": 0.923
    },
    "recall": {
      "p2.5": -0.051470588235294046,
      "p50": -0.007407407407407363,
      "p97.5": 0.03875968992248058,
      "share_below_0": 0.557
    },
    "fpr": {
      "p2.5": -0.05358349420849418,
      "p50": -0.009478885893980216,
      "p97.5": 0.03810439560439562,
      "share_below_0": 0.58
    }
  },
  "folds_with_lower_recall_gap": "2/3",
  "checks": {
    "recall_gap_reduction_ci_below_0": false,
    "pooled_recall_meets_gate": true,
    "fpr_increase_within_limit": true,
    "lower_gap_in_all_folds": false
  },
  "validated": false
}
```